

The questions are selected by the class to review.

Homework 5 Problem 4. Like group homomorphism, there are special functions connecting rings.

Definition. Let R, S be commutative rings. A function $f : R \rightarrow S$ is a ring homomorphism if for all $a, b \in R$

$$f(1_R) = 1_S, \quad (a + b) = f(a) + f(b), \quad f(ab) = f(a)f(b).$$

Let $f : R \rightarrow S$ be a ring homomorphism. Show the following.

1. Show $\text{im}(f) = \{s \in S : s = f(r) \text{ for some } r \in R\}$ is a subring and $\ker(f) = \{r \in R : f(r) = 0\}$ is an ideal.
2. Let $I \subseteq S$ be an ideal. Show $f^{-1}(I) = \{r \in R : f(r) \in I\}$ is also an ideal.
3. Suppose f is surjective. Let $J \subseteq R$ be an ideal. Then $f(J)$ is an ideal.
4. Give a ring homomorphism $f : R \rightarrow S$ such that $f(J)$ is not an ideal while J is an ideal.
5. (Just Think about it.) Define $\rho : \mathbb{Z} \rightarrow \mathbb{Z}$ by $n \mapsto n + 1$. Note $\rho^{-1}(2\mathbb{Z}) = 2\mathbb{Z} + 1$. Does this example contradicts the result in part 2?

Proof. 1. We show $\ker f$ is an ideal. First, $0 \in \ker f$ because $f(0) = 0$ as f is a ring homomorphism. Second, let $x, y \in \ker f$. Then $f(x + y) = f(x) + f(y) = 0 + 0 = 0$. So $x + y \in \ker f$. Finally let $x \in \ker f$ and $r \in R$. Then $f(rx) = f(r)f(x) = f(r) \cdot 0 = 0$. So $rx \in \ker f$.

2. First, because I is an ideal, $0 \in I$. Then $f(0) = 0 \in I$ implies $0 \in f^{-1}(I)$. Second, let $x, y \in f^{-1}(I)$. Then $f(x + y) = f(x) + f(y) \in I$ since I is closed under addition. So $x + y \in f^{-1}(I)$. Finally let $x \in f^{-1}(I)$ and $r \in R$. Then $f(rx) = f(r)f(x) \in I$ since $f(x) \in I$ and I is closed with respect to multiplication by ring element. So $rx \in f^{-1}(I)$.
3. Again since $0 \in J$, we have $0 = f(0) \in f(J)$. Second, let $x, y \in f(J)$. We can write $x = f(x')$ and $y = f(y')$ for $x', y' \in J$. Then $x + y = f(x') + f(y') = f(x' + y') \in f(J)$ as $x' + y' \in J$. Finally, let $x = f(x') \in f(J)$ and $s \in S$. Then we may write $s = f(r)$ for some $r \in R$. Now $sx = f(r)f(x') = f(rx') \in f(J)$ as $rx' \in J$.
4. By part 3, we must consider a nonsurjective homomorphism. For instance, the inclusion $\mathbb{Z} \rightarrow \mathbb{Q}$ does not take $2\mathbb{Z}$ to an ideal in $\mathbb{Q}$. This is because $\mathbb{Q}$ is a field and has two ideals (0) and $\mathbb{Q}$. Clearly $2\mathbb{Z}$ is neither. (Marcial.) The inclusion $\mathbb{Z} \rightarrow \mathbb{Z}[x]$ does not take $2\mathbb{Z}$ to an ideal in $\mathbb{Z}[x]$. This is because any nonzero ideal in $\mathbb{Z}[x]$ will contain a polynomial of positive degree.
5. (Nikoli.) There is no contradiction because ρ is not a ring homomorphism. Note that $\rho(1) = 2$ but $\rho(1)\rho(1) = 4$. In particular, ρ is only an group homomorphism with respect to addition. $\square$

Homework 6 Problem 5. Let k be a field. Consider the polynomial ring over the polynomial ring over a field

$$k[x][y] = \left\{ \sum_{i=0}^n f_i(x)y^i : f_i(x) \in k[x], n \in \mathbb{N} \right\}.$$

This is really just $k[x, y] = \left\{ \sum_{i=0}^n f_i(x)y^i : f_i(x) \in k[x] \right\}$. Explain why $\mathbb{Z}[x]$ is not an ED by showing (x, y) is not a principle ideal.

Proof. Suppose (x, y) is a principle ideal and say $(x, y) = (f)$ for $f = f(x, y) \in k[x, y]$. Note x, y are both irreducible in $k[x, y]$. Indeed if $x = g(x, y)h(x, y)$, then both g and h contains no powers of y and we are back in the setting of $k[x]$. So $f \mid x$ and $f \mid y$ implies $f \in k$ which implies $(x, y) = (f) = k[x, y]$. This is not true since $2 \notin (x, y)$. $\square$

Homework 7 Problem 4. Let R be a commutative ring and $p \in R$. Define the evaluation map $ev_p : R[x] \rightarrow R$ by $f(x) \mapsto f(p)$, which is a ring homomorphism. We say p is a root of $f(x) \in R[x]$ if $ev_p(f(x)) = f(p) = 0$ in R .

1. Show p is a root of $f(x)$ iff $f(x) = (x - p)g(x)$ for some $g(x) \in R[x]$.
2. What is the kernel of ev_p in $R[x]$?
3. Show that when R is a domain, then $(x - p)$ is a prime ideal in $R[x]$.

Proof. 1. ($\implies$) Suppose p is a root and write $f(x) = \sum_{i=0}^n a_i x^i$. Then $f(p) = 0$ and $f(x) - f(p) = f(x)$. Note we can write the LRS as

$$f(x) - f(p) = \sum_{i=0}^n a_i (x^i - p^i)$$

and $x^n - p^n = (x - p)(x^{n-1} + x^{n-2}p + \dots + p^{n-1})$. So $x - p$ divides each term in $f(x) - f(p)$.

($\impliedby$) This direction is easy because evaluation map is a ring homomorphism. So $ev_p(f) = ev_p((x - p)g(x)) = ev_p((x - p))ev_p(g(x)) = 0 \cdot g(p) = 0$.

2. It is $(x - p)$ as p is a root of $f(x)$ iff $f(x) \in \ker ev_p$ by definition. Then it follows from part 1.
3. Suppose $fg \in (x - p)$. Then $f(p)g(p) = 0$ in R which is domain iff $f(p) = 0$ or $g(p) = 0$ iff $f(p) \in \ker ev_p$ or $g(p) \in \ker ev_p$. $\square$